25/7/26

NAME: PARVEEN
REG.NO: 192462005

UNIT: I

ANALYTICAL QUESTIONS

1. • Key = 3
   • Plaintext = MEET ME AFTER THE TOGA PARTY
   • Found by trying all 26 keys (brute force.)
2. • Cipher text : DPNFCTEJ
   • Formula C: (P+11) % 26
3. • Only 26 keys
   • Easy to try all keys (brute - force attack)
4. • Letters keek same frequency.
   • Attackers use frequency analysis to guess letters
5. • Cipher : Atbash cipher.
   • Plaintext : THIS IS A SECRET
6. • Assume X = E
   • Then test other common letters & words
7. • One letter can become different ciphertext letters
   • Frequency analysis becomes harder.
8. • Ciphertext = MVWZXMQVYZLW
   • Use key repeatedly for encryption.
9. • Plaintext = ATACKAT DAWN
10. • Repeated key creates katterns.
    • Kasiski attack finds key length & helps break ciph

11. • It becomes a One-Time Pad.
• It is unbreakable if key is truly random & use only once.

12. • Monoalphabetic = One fixed substitution less secure.
• Polyalphabetic = Multiple substitution more secure.

13. • Letters are treated as numbers (A=0, B=1, . . . Z=25)
• Encryption & decryption use mod 26.

14. • Plaintext = HI
• Ciphertext = TC

15.
• If key matrix has no inverse, decryption is not possible.

16.
• So key matrix has inverse.
• Without it decryption fails.

17.
• It encrypts blocks of letters using matrix multiplication.

18.
• It is likely a polyalphabetic substitution cipher.
• Frequency analysis becomes difficult.

19. • Use random letter mapping.
• Share key through secure / private channel.

20. • Yes, ciphertext blocks will also be same if same key is used.

21.
• No
• It is Affine cipher, not Caesar (shift) cipher

22.
- The attacker knows some plaintext & its ciphertext,
- Hence find encryption key.

23.
- It changes letters using different shifts
- But 1 plaintext letter affects only 1 ciphertext.

24.
- "THE" is very common in English.
- Attackers match patterns to guess letters substitution

25.
- Caesar Cipher
- Monoalphabetic Cipher
- Vigenere Cipher
- Hill Cipher

26.
- Verify zig-zag placement of letters.
- If ciphertext doesn't match, rail traversal was done incorrectly.

27.
- Fill table row by row.
- Add padding if last row is incomplete.

28.
- Repeated letters make column numbering confusing
- Use fixed rule to order equal letters.

29.
- Verify that letters follow correct zig-zag patt
- If not, cipher text is incorrect.

30.
- Possible matrix sizes depend on message length

31. • Reconstruct columns using key order.
• Rearrange columns to find plaintext.

32. • Reconstruct columns using key order.
• Rearrange columns to find plaintext.

33. • This can happen if keys produce same column order.

34. • Try 2, 3, 4, ... rails.
• Choose result that gives meaningful english text.

35. • More key permutations
• Brute - force attack becomes much harder.

36. • Attackers can guess padding.
• This may reveal message structure.

37. • Matrix size options are limited.
• Padding is usually required.

38. • Decryption gives meaningless text.
• Correct row/column order is essential.

39. • They stay on first & last rails.
• So they often remain in same position.

40. • Total cells = 24 (6×4)
• 1 cell is left empty.

41. • Each 5 - letter key 5! = 120
Combinations = 120 × 120 = 14400

42. • Try single trasposition first.
• If it fails, try double.

43. • Repeated letters make column ranking confusing.

44. • It usually indicates columnar Transposition.
- Rail Fence work slow.

45. • It suggest no. of rails was chosen so letters are evenly spread.

46. • This is slighly weaken cipher.
• May create patterns

47. • Padding is added to fill empty cells.
• so ciphertext length increases.

48. • 6! = 720
• By - brute force approach

49. • Compare result using 2 rails & 3 rails.
• Correct rail no. is CRYPTOGRAPHY.

50. • Numeric Key: Easy to use
• Keyword keys: Easy to remember.